

Report conversation

In older Fortran codes, one often uses the following syntax when passing an array into a subroutine "f" that expects an array:

```
call f(A(10))
```

And the meaning of this syntax is that it passes an array section A(10:), that is, it is a pointer to element number 10, and inside the subroutine "f", it behaves like an array.

Yes, you are correct. In older Fortran codes, particularly in versions prior to Fortran 90, this syntax was commonly used to pass array sections to subroutines.

When you use `call f(A(10))`, you are passing a section of the array `A` starting from element 10 onwards. This means that inside the subroutine `f`, the array will start from element 10, but the subroutine itself is unaware of the original size of the array `A`.

It's important to note that this approach had some limitations and could lead to potential issues if not used carefully. For example, if the subroutine `f` assumes that the array passed to it always starts from element 1, it may not behave as expected.

With the introduction of Fortran 90 and later versions, array syntax and handling was significantly improved with the inclusion of explicit array dimensions and dynamic memory allocation, making it easier and more intuitive to work with arrays in Fortran. This allowed for more flexible and robust handling of arrays in subroutines.

We are developing a Fortran compiler. We do not want this feature to be allowed by default, only with a compiler option. What would be a good name of such an option?

A good name for an option that enables this older Fortran array section behavior could be:

`-flegacy-array-sections`

This name clearly indicates that it's related to the legacy behavior of array sections in Fortran. Users who are familiar with this feature from older Fortran versions would likely recognize its purpose from this option name.

Very good. Give me 5 other ideas for such a name.

Certainly! Here are five alternative names for the compiler option:

1. `-fallow-legacy-arrays`
2. `-fcompat-legacy-array`
3. `-fenable-legacy-array`
4. `-fbackwards-array`
5. `-fretro-array-sections`

These options all convey the idea that they enable a legacy or older style of array behavior in Fortran.

Can you give me 10 more ideas?

Certainly! Here are ten more ideas for the compiler option:

6. `-fclassic-array-sections`
7. `-ftraditional-array-passing`
8. `-fbackwards-compatible-arrays`
9. `-fhistorical-array-syntax`
10. `-fretrofitted-array-sections`
11. `-flegacy-array-syntax`
12. `-fcompat-array-passing`
13. `-fallow-historical-arrays`

14. `-fenable-old-array-style`

15. `-fretrofit`